

ICT703 Programming

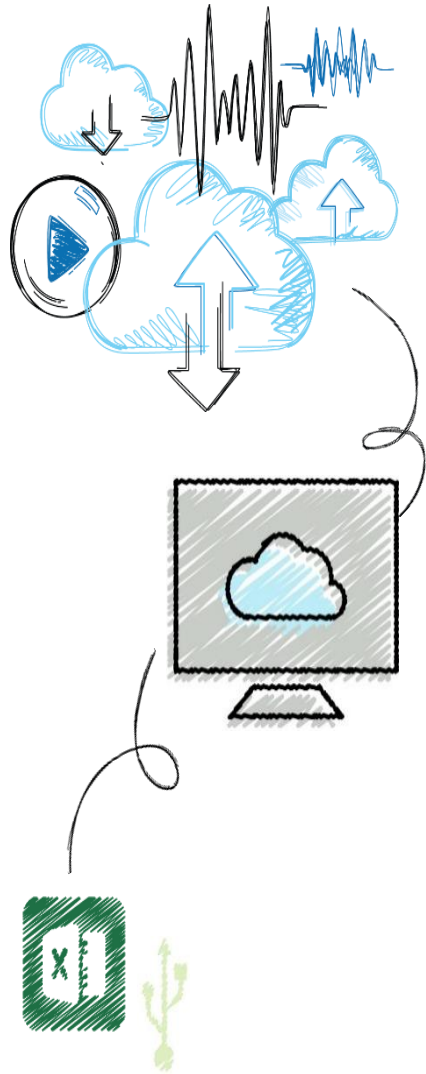
Module 6

Sets



Objectives

1. Describe a Set Data Type and its uses
2. Combining Sets





Programming Concepts

Four control flow

- Sequence
- Iteration
- Selection
- Abstraction – methods/classes

Data



Introduction

- A **set** resembles a list, but is immutable and do not allow duplicate values
 - `fruit = {"apple", "banana", "cherry"}`
 - A set is a collection which is unordered, unchangeable, and unindexed.
 - Unordered - the items in a set do not have a defined order and cannot be referred to by index or key.
 - Unchangeable - we cannot change the items after the set has been created, but you can remove items and add new items.
 - Duplicates Not Allowed - sets cannot have two items with the same value.
 - `print({"apple", "banana", "cherry", "apple"})`
 - `{'apple', 'banana', 'cherry'}`
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Set - Methods

- We can add elements to a set
 - `fruit = {"apple", "banana", "cherry"}`
 - `fruit.add("orange")`
 - We can remove elements to a set
 - `fruit.remove("banana")`
 - We can join 2 sets. The result has no duplicates
 - `fruit = {"apple", "banana", "cherry"}`
 - `fruit.union({"banana", "cherry", "pineapple"})`
 - `{"apple", "banana", "cherry", "pineapple"}`
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Sets are good to keep unique lists

- Create a list of student first names and continually add new names.
 - `firstNames = {'Andrew', 'Fred', 'Peter', 'Jane'}`
 - `newPerson = input("What is your name ")`
 - `while newPerson != "stop":`
 - `firstNames.add(newPerson)`
 - `newPerson = input("What is your name ")`
 - `print(firstNames)`
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Summary

- A set is a collection which is unordered, unchangeable, and unindexed.
- They are useful when you want to keep a list unique